Basin parameters

SWAT+ Documentation

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Purpose

`codes.bsn` holds the method-selection switches that apply to the whole basin: the PET method, the routing method, the carbon model, the soil P model, and so on. It is read into `bsn_cc` (`type basin_control_codes`) early in the run and then consulted by every process module.

Source

- Reader: src/basin_read_cc.f90
- Type definition: src/basin_module.f90, `type basin_control_codes`

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Line 3: 26 fields, list-directed, read into `bsn_cc` in declaration order.

#	Field	Type	Default	Description
1	petfile	char(16)	`pet.cli`	Potential ET filename. Used when `pet = 3` (read PET from file).
2	wwqfile	char(16)	`""`	Watershed stream water-quality filename.
3	pet	int	0	PET method. `0` Priestley-Taylor, `1` Penman-Monteith, `2` Hargreaves. (Method `3` triggers reading `pet.cli`.)
4	nam1	int	0	Not used.
5	crk	int	0	Crack flow. `1` to compute flow in cracks.
6	swift_out	int	0	SWIFT output. `0` do not write, `1` write `swift_hru.inp`.
7	sed_det	int	0	Peak rate method. `0` NRCS dimensionless hydrograph with PRF, `1` half-hour rainfall intensity method.
8	rte	int	0	Water routing method. `0` variable storage, `1` Muskingum.
9	deg	int	0	Not used.
10	wq	int	0	Not used.
11	nostress	int	0	Plant stress. `0` all stresses applied, `1` turn off all plant stress, `2` turn off nutrient stress only.
12	cn	int	0	Not used.
13	cfac	int	0	Not used.
14	cswat	int	0	Carbon model. `0` static (legacy mineralization), `1` C-FARM (reserved, not implemented), `2` dynamic CENTURY/SWAT-C model.
15	lapse	int	0	Lapse-rate adjustment for precip and temperature. `0` no, `1` yes.
16	uhyd	int	1	Unit hydrograph. `0` triangular, `1` gamma function.
17	sed_ch	int	0	Not used.
18	tdrn	int	0	Tile drainage equation. `0` drawdown-days, `1` DRAINMOD.
19	wtdn	int	0	Shallow water-table depth. `0` original routine (fill to upper limit), `1` DRAINMOD water-table depth.
20	sol_p_model	int	0	Soil P model. `0` original SWAT P model, `1` Vadas and White (2010).
21	gampt	int	0	Infiltration method. `0` curve number, `1` Green-Ampt.
22	atmo	char(1)	`a`	Not used.
23	smax	int	0	Not used.
24	qual2e	int	0	Instream nutrient routing. `0` QUAL2E, `1` QUAL2E with simplified transformations.
25	gwflow	int	0	gwflow module. `0` inactive, `1` active. Triggers extra logic in `object.cnt`.
26	idc_till	int	3	Tillage method when `cswat = 2`. `1` DSSAT, `2` EPIC, `3` Kemanian, `4` DNDC.

The "Not used" fields are read from the file but the model does not consult them. Tools commonly write `0`.

Side effects after read

If `pet = 3`, the reader opens `pet.cli` and reads a title and header line to position the file pointer. The actual PET values are read later by the climate routines.

Example

`refdata/Ames_sub1/codes.bsn`:

codes.bsn:

pet_file	wq_file	pet	event	crack	swift_out	sed_det	rte_cha	deg_cha
null	null	2	0	0	0	0	0	0

The header column names written by SWAT+ Editor (`event`, `rte_cha`, `i_fpwet`, …) differ from the field names in the source (`crk`, `rte`, `qual2e`, …). Order is what the reader cares about. Mapping the example to the table:

- `pet = 2`: Hargreaves.
- `cswat = 2`: dynamic CENTURY/SWAT-C carbon model. (Header label `carbon`.) The example file above still carries `carbon = 1`, which is now the reserved C-FARM slot and runs no carbon model; datasets that want the dynamic model must set `carbon = 2`.
- `uhyd = 1`: gamma-function unit hydrograph.
- `idc_till = 3`: Kemanian tillage method.

The `null` strings in columns 1 and 2 disable the PET and water-quality file inputs.

Related files

- `parameters.bsn`. Numeric basin parameters consumed by the methods selected here.
- `object.cnt`. Reads gwflow state when `gwflow = 1`.
- `pet.cli`. Used when `pet = 3`.

Purpose

`parameters.bsn` holds the basin-wide numeric coefficients (initial soil water, surface runoff lag, nutrient cycling rates, channel routing factors, lapse rates, CO2, and so on). Values are read into `bsn_prm` (`type basin_parms`) and consulted by hydrology, nutrient, erosion, and routing routines.

Source

- Reader: `src/basin_read_prm.f90`
- Type definition: `src/basin_module.f90`, `type basin_parms`

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Line 3: 44 values, list-directed, read into `bsn_prm` in declaration order. The last two values are integers; the rest are reals.

#	Field	Type	Default	Units	Description
1	evlai	real	3.0	none	leaf area index above which no soil evaporation occurs
2	ffcb	real	0.0	fraction	initial soil water content as a fraction of field capacity
3	surlag	real	4.0	days	surface runoff lag time
4	adj_pkr	real	1.0	none	peak rate adjustment factor in the subbasin
5	prf	real	484.0	none	peak rate factor for the peak rate equation
6	spcon	real	0.0	none	not used
7	spexp	real	0.0	none	not used
8	cmn	real	0.003	1/day	rate factor for mineralization of active organic N
9	n_updis	real	20.0	none	nitrogen uptake distribution parameter
10	p_updis	real	20.0	none	phosphorus uptake distribution parameter
11	nperco	real	0.10	0..1	nitrate percolation coefficient
12	pperco	real	10.0	0..1	phosphorus percolation coefficient
13	phoskd	real	175.0	none	phosphorus soil partitioning coefficient
14	psp	real	0.40	none	phosphorus availability index
15	rsdco	real	0.05	none	residue decomposition coefficient
16	percop	real	0.5	0..1	pesticide percolation coefficient
17	msk_co1	real	0.75	none	Muskingum storage time constant for bankfull depth
18	msk_co2	real	0.25	none	Muskingum storage time constant for low flow
19	msk_x	real	0.20	none	Muskingum weighting factor for inflow vs outflow
20	nperco_lchtile	real	0.5	none	N concentration coefficient for tile flow and bottom-layer leaching
21	evrch	real	0.60	none	reach evaporation adjustment factor
22	scoef	real	1.0	0..1	channel storage coefficient
23	cdn	real	1.40	none	denitrification exponential rate coefficient
24	sdnco	real	1.30	none	denitrification threshold (fraction of field capacity)
25	bact_swf	real	0.15	fraction	fraction of manure containing active colony-forming units
26	tb_adj	real	0.0	none	adjustment factor for sub-daily unit hydrograph base time
27	cn_froz	real	0.000862	none	parameter for frozen-soil adjustment on infiltration/runoff
28	dorm_hr	real	-1.0	hours	dormancy time threshold
29	plaps	real	0.0	mm/km	precipitation lapse rate per km of elevation difference
30	tlaps	real	6.5	deg C/km	temperature lapse rate per km of elevation difference
31	nfixmx	real	20.0	kg/ha/day	maximum daily nitrogen fixation
32	decr_min	real	0.01	1/day	minimum daily residue decay
33	rsd_covco	real	0.75	none	residue cover factor for computing cover fraction
34	urb_init_abst	real	1.0	none	maximum initial abstraction for urban areas (Green-Ampt)

#	Field	Type	Default	Units	Description
35	petco_pmpt	real	100.0	percent	PET adjustment for Penman-Monteith and Priestley-Taylor
36	uhalpha	real	1.0	none	alpha coefficient for the gamma-function unit hydrograph
37	eros_spl	real	0.0	none	coefficient of splash erosion (typical range 0.9 to 3.1)
38	rill_mult	real	0.0	none	rill erosion coefficient
39	eros_expo	real	0.0	none	exponential coefficient for overland flow
40	c_factor	real	0.0	none	cover and management factor scaling for overland-flow erosion
41	ch_d50	real	0.0	mm	median particle diameter of the main channel
42	co2	real	400.0	ppm	CO2 concentration at start of simulation
43	day_lag_mx	int	0	days	maximum days to lag hydrographs for HRU, RU, and channels on non-draining soils
44	igen	int	5	none	random generator code. `0` use defaults, `1` generate new numbers every run

Example

refdata/Ames_sub1/parameters.bsn:

```
parameters.bsn: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
  lai_noevap       sw_init      surq_lag      adj_pkrt  adj_pkrt_sed       lin_sed       exp_sed      orgn
     3.00000       0.50000       4.00000       1.00000       1.00000       0.00000       1.00000       0.0
```

Notes:

- Tool header names (`lai_noevap`, `sw_init`, `surq_lag`, `adj_pkrt`, `orgn_min`, `n_uptake`, `evap_adj`, `denit_exp`, `denit_frac`, `man_bact`, `adj_uhyd`, `n_fix_max`, `rsd_decay`, `rsd_cover`, `uhyd_alpha`, `splash`, `rill`, `surq_exp`, `cov_mgt`, `cha_d50`, `day_lag_max`) do not match the field names in the source (`evlai`, `ffcb`, `surlag`, `adj_pkr`, `cmn`, `n_updis`, `evrch`, `cdn`, `sdnco`, `bact_swf`, `tb_adj`, `nfixmx`, `decr_min`, `rsd_covco`, `uhalpha`, `eros_spl`, `rill_mult`, `eros_expo`, `c_factor`, `ch_d50`, `day_lag_mx`). The reader uses position only.
- Position 8 in this dataset is `orgn_min = 0.0`. In the source this is `cmn`, with default `0.003`. The example overrides the default to zero, which disables active organic-N mineralization.
- Position 7 is `exp_sed = 1.0`. This is `spexp` in the source, marked "not used".
- Positions 28 to 30 (`dorm_hr`, `plaps`, `tlaps`) are all zero in this dataset. The default for `tlaps` is `6.5` deg C/km; this dataset disables temperature lapse.

Related files

- codes.bsn. Method switches that select which of these parameters are active.

Purpose

`carbon.bsn` holds the basin-wide scalar coefficients for the SWAT+ carbon module. The reader fills `org_frac`, `cb_wtr_coef`, `man_coef`, and `org_con` (all defined in carbon_module.f90), the basin-scope tillage variables in tillage_data_module.f90 (`bio_consf`, `till_consf`, `bmix_a/b/c`, `tillmix_a/b/c`, `till_eff_days`), the `photo_degrade_factor` in plant_data_module.f90, and the five residue-decomposition tunables (`n_act_frac`, `cnr_cap`, `cnr_ref`, `cpr_cap`, `cpr_ref`) promoted from hard-coded constants into module-scope reals in `carbon_module.f90`.

`carbon.bsn` replaces the legacy `carb_coefs.cbn` (keyword-style) and works alongside the per-layer file carbon_lyr.bsn, which holds the rate constants and CO2 fractions that differ between the surface and subsurface layer groups.

Source

- Reader: src/carbon_bsn_read.f90
- Filename slot: src/input_file_module.f90, `type input_basin`, field `carbon_bsn = "carbon.bsn"`.
- Target types: `organic_fractions`, `carbon_water_coef`, `manure_coef`, `organic_controls` in src/carbon_module.f90; module-scope reals `n_act_frac`, `cnr_cap`, `cnr_ref`, `cpr_cap`, `cpr_ref` in the same module; `bio_consf`, `till_consf`, `till_eff_days`, `bmix_a/b/c`, `tillmix_a/b/c` in src/tillage_data_module.f90; `photo_degrade_factor` in src/plant_data_module.f90.

Activation

The reader runs only when `bsn_cc%cswat == 2` (set in codes.bsn). When `cswat == 2` the file is **required**: if `carbon.bsn` (or its companion `carbon_lyr.bsn`) is missing or its data row cannot be parsed, the reader writes an error message to stdout and unit 9001 and calls `error stop`. There is no silent fallback to compiled defaults. `carbon_bsn_read.f90` returns early for any other `cswat` value, so `cswat = 1` (the reserved C-FARM slot) does not read this file.

The dynamic carbon model activates on `cswat = 2`, matching legacy SWAT numbering (1 = C-FARM, 2 = CENTURY). It was previously `cswat = 1`; existing `codes.bsn` files left at `carbon = 1` now run with the carbon model off and must be changed to `carbon = 2`.

`carbon.bsn` is listed in the BASIN row of `file.cio` alongside `codes.bsn` and `parameters.bsn` (three columns). `carbon_lyr.bsn` rides at a fixed name beside it and does not appear in `file.cio`.

> **Note**
>
> The legacy `cbn_diag` first column is retired. Output gating now lives entirely in `print.prt` through the `hru_cb_*` and `lsu_cb_*` rows (see the carbon outputs reference).

Format

- Line 1: title (skipped).
- Line 2: column header (skipped).
- Line 3: 28 values, list-directed, read into the target fields in the order below. (The 28th value, `mathers_method`, was added when carbonDev merged to main; 27-value files from before the merge fail to parse and must add the trailing column.)

Values are space-delimited. Position is significant; the column header is decorative.

#	Field	Type	Target	Units	Description
1	init_seq	real	org_frac%frac_seq	fraction	initial fraction of total soil carbon held in the sequestered pool
2	init_microb	real	org_frac%frac_hum_microb	fraction	initial microbial-humus fraction of non-sequestered carbon
3	init_slow	real	org_frac%frac_hum_slow	fraction	initial slow-humus fraction of non-sequestered carbon
4	init_passive	real	org_frac%frac_hum_passive	fraction	initial passive-humus fraction of non-sequestered carbon
5	koc_c	real	cb_wtr_coef%prmt_21	none	partition coefficient for carbon loss in water and sediment (formerly prmt_21)
6	solc_ratio	real	cb_wtr_coef%prmt_44	0..1	ratio of soluble C in surface runoff to soluble C in percolate (formerly prmt_44)
7	till_eff_days	real	till_eff_days	days	number of days a tillage event remains effective
8	manure_c_frac	real	man_coef%rtof	none	manure organic N and P partition between fresh and stable pools (formerly rtof)
9	bio_consol	real	bio_consf	none	biological consolidation factor for soil moisture mixing
10	till_consol	real	till_consf	none	tillage consolidation factor for soil moisture mixing
11	tmpf_eqn	int	org_con%tmpf	none	temperature factor selector (1, 2, or 3) used in cbn_zhang2.f90
12	watf_eqn	int	org_con%watf	none	water factor selector (1 or 2) used in cbn_zhang2.f90
13	t_cbn_min	real	org_con%tn	deg C	minimum temperature bound for tmpf2 (lower limit of biological activity)
14	t_cbn_opt	real	org_con%top	deg C	optimum temperature for tmpf2
15	t_cbn_max	real	org_con%tx	deg C	maximum temperature bound for tmpf2 (upper limit of biological activity)
16	bmix_a	real	bmix_a	none	biological mixing curve coefficient A (mgt_tillfactor.f90 tillf option 4)
17	bmix_b	real	bmix_b	none	biological mixing curve coefficient B
18	bmix_c	real	bmix_c	none	biological mixing curve coefficient C
19	tillmix_a	real	tillmix_a	none	tillage mixing curve coefficient A
20	tillmix_b	real	tillmix_b	none	tillage mixing curve coefficient B
21	tillmix_c	real	tillmix_c	none	tillage mixing curve coefficient C
22	sfc_rsd_photodeg	real	photo_degrade_factor	1/day	surface residue photodegradation rate factor
23	n_act_frac	real	n_act_frac	fraction	fraction of organic N held in the active humus pool (drives active-to-stable flow in nut_nminrl.f90)
24	cnr_cap	real	cnr_cap	none	upper cap on residue C:N ratio before computing the decomposition factor
25	cnr_ref	real	cnr_ref	none	reference C:N ratio where the C:N decomposition factor equals 1
26	cpr_cap	real	cpr_cap	none	upper cap on residue C:P ratio before computing the decomposition factor
27	cpr_ref	real	cpr_ref	none	reference C:P ratio where the C:P decomposition factor equals 1

#	Field	Type	Target	Units	Description
28	`mathers_method`	int	`org_frac%mathers_method`	flag	slow-humus pool initialization: `0` original humus-slow init (default), `1` Mathers method (clay+silt based, from “Updating carbon pool initialization with DSSAT-CENTURY”). Used in `soil_nutcarb_init`.

The reader does not interpret column 1-4 as a probability simplex; the four `init_*` values are used as supplied. The four target fields hold the fractional partitioning of soil carbon into sequestered, microbial-humus, slow-humus, and passive-humus pools at simulation start.

The `tmpf_eqn` (col 11) and `watf_eqn` (col 12) integers select between the temperature and water response equations encoded in `cbn_zhang2.f90`. They are integers in the file but the reader does not enforce that; the consuming code treats them as `int(real)`.

Example

`refdata/Lafayette/carbon.bsn`:

```
carbon.bsn: basin-wide scalar carbon parameters (one record)
   init_seq   init_microb   init_slow   init_passive   koc_c   solc_ratio   till_eff_days   manure_c_frac
       0.95         0.02         0.44           0.54    1000          0.5             100            0.50          0.1
```

Calibration

Eighteen of the 28 fields are wired in `cal_parm_select.f90` and may be touched through `calibration.cal`: `init_seq`, `init_microb`, `init_slow`, `init_passive`, `koc_c`, `solc_ratio`, `manure_c_frac`, `bio_consol`, `till_consol`, `sfc_rsd_photodeg`, `n_act_frac`, `cnr_cap`, `cnr_ref`, `cpr_cap`, `cpr_ref`, `t_cbn_min`, `t_cbn_opt`, `t_cbn_max`. The six bmix/tillmix curve coefficients are also wired. The selector integers (`tmpf_eqn`, `watf_eqn`) and the integer-valued `till_eff_days` are not calibrated.

> **Note**
>
> `proc_cal` must run **before** `soil_nutcarb_init` for the `init_*` calibration values to take effect. The current `main.f90` already orders these correctly; do not reorder.

Related files

- `carbon_lyr.bsn`. Per-layer rate constants and CO2 allocation fractions.
- `codes.bsn`. The `cswat` switch that gates this file.
- `print.prt`. Output gating for the carbon families.
- Soil carbon (model theory). Equations and pool structure.
- Parameter change files. Calibration entries for the wired carbon parameters.

Purpose

`carbon_lyr.bsn` holds the carbon coefficients that differ between soil layer groups. It populates `carbdb(layer_id)` (rate constants for the litter, microbial, slow, and passive pools) and `org_allo(layer_id)` (the CO2 fractions produced by each pool transformation). The two layer groups currently in use are:

- `layer 1` (surface litter and topsoil)
- `layer 2` (all subsurface layers)

The reader indexes directly into the `carbdb(:)` and `org_allo(:)` arrays. Both are sized `dimension(2)` in `carbon_module.f90` today; a future expansion to additional layer groups requires only a larger array allocation, not a reader change.

Source

- Reader: same routine as `carbon.bsn`, `src/carbon_bsn_read.f90`. The per-layer block is parsed in the second half of the subroutine.
- Target types: `carbon_inputs` (`carbdb`) and `organic_allocations` (`org_allo`) in `src/carbon_module.f90`.
- Filename: hardcoded in the reader as `carbon_lyr.bsn`. **Not listed in `file.cio`** — it rides at a fixed name beside `carbon.bsn`.

Activation

Read only when `bsn_cc%cswat == 2` (the dynamic CENTURY/SWAT-C model; the activation value moved from 1 to 2). Like `carbon.bsn`, the file is **required**: an absent file, an unparseable row, or fewer data rows than the size of `carbdb` (currently 2) triggers `error stop` with a clear message.

Format

- Line 1: title (skipped).
- Line 2: column header (skipped).
- Lines 3..: one row per layer group. Each row begins with an integer `layer` id (1 = top, 2 = subsurface) followed by 11 real coefficients. The reader loops until end-of-file. Rows with an out-of-range `layer` id (less than 1 or greater than `size(carbdb)`) are skipped with a warning to unit 9001 and do not count toward the required row count.

#	Field	Type	Target	Units	Description
1	`layer`	int	(array index)	none	layer group id; 1 = top (surface litter + topsoil), 2 = all subsurface layers
2	`hp_rate`	real	`carbdb(layer)%hp_rate`	1/day	passive humus transformation rate under optimal conditions
3	`hs_rate`	real	`carbdb(layer)%hs_rate`	1/day	slow humus transformation rate under optimal conditions
4	`microb_rate`	real	`carbdb(layer)%microb_rate`	1/day	microbial biomass transformation rate under optimal conditions
5	`meta_rate`	real	`carbdb(layer)%meta_rate`	1/day	metabolic litter transformation rate under optimal conditions
6	`str_rate`	real	`carbdb(layer)%str_rate`	1/day	structural litter transformation rate under optimal conditions
7	`microb_top_rate`	real	`carbdb(layer)%microb_top_rate`	1/day	top-layer microbial activity adjustment
8	`hs_hp`	real	`carbdb(layer)%hs_hp`	none	slow-to-passive humus allocation coefficient
9	`a1co2`	real	`org_allo(layer)%a1co2`	fraction	decomposed metabolic litter routed to CO2
10	`asco2`	real	`org_allo(layer)%asco2`	fraction	decomposed slow humus routed to CO2
11	`apco2`	real	`org_allo(layer)%apco2`	fraction	decomposed passive humus routed to CO2
12	`abco2`	real	`org_allo(layer)%abco2`	fraction	decomposed microbial biomass routed to CO2

Example

`refdata/Lafayette/carbon_lyr.bsn`:

```
carbon_lyr.bsn: layer-keyed carbon coefficients (one row per layer group; 1 = top, 2 = subsurface; extend
   layer       hp_rate       hs_rate   microb_rate   meta_rate   str_rate   microb_top_rate   hs_hp   a1co
       1      1.2e-05      2.92e-04        0.0164      0.0405     0.0107            0.0164    0.05   0.60
       2      1.2e-05      1.81e-04          0.02      0.0507     0.0134              0.02    0.05   0.55
```

Calibration

All 11 per-layer fields are wired in `cal_parm_select.f90`. The calibration target is the `carbdb(ly)` or `org_allo(ly)` element selected by the `ly` index in the calibration record (see Parameter change files):

- `hp_rate`, `hs_rate`, `microb_rate`, `meta_rate`, `str_rate`, `microb_top_rate`, `hs_hp` target `carbdb(ly)%*`.
- `a1co2`, `asco2`, `apco2`, `abco2` target `org_allo(ly)%*`.

Each `case` branch in `cal_parm_select.f90` guards `ly` to the range `[1, size(carbdb)]` (or `[1, size(org_allo)]`), so out-of-range layer indices in a calibration row are silently no-op rather than crashing.

Related files

- `carbon.bsn`. Basin-wide scalar carbon parameters.
- `codes.bsn`. The `cswat` switch.
- Soil carbon (model theory). Pool definitions and Century-style transformation equations.